Chapter-1

Introduction to Computer System

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1. Computer System

Definition

A **computer system** is an electronic device that accepts data as input, processes it according to a set of instructions (program), stores the processed information, and produces meaningful output.

A computer works based on the **Input → Process → Output (IPO)** cycle.

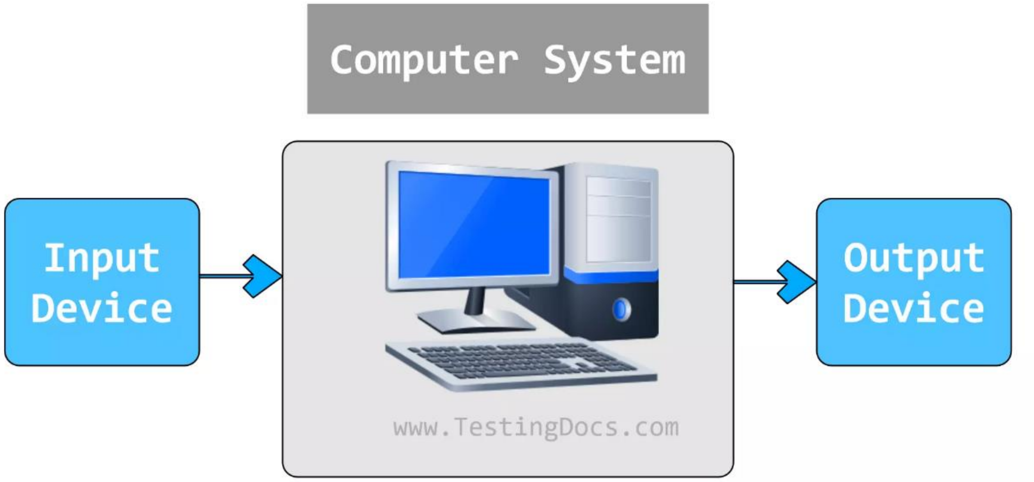

Fig 1.a) Structured Diagram of a Computer

Computer input and output (I/O) units are hardware components that enable communication between the user and the computer. Input devices are used to enter data and instructions into the computer, while output devices present the processed information to the user in the form of text, images, audio, or printed documents.

1.1 Invention of the computer

The invention of the computer was not the work of a single person. It evolved over many years through the contributions of several inventors. The foundation of the modern computer was laid by **Charles Babbage (1837),** who designed the Difference Engine and the Analytical Engine.

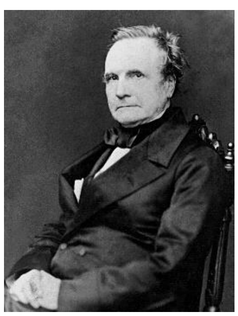

Fig 1.1a) Father of the Computer

1.2 Generations of Computers

1. First Generation (1940–1956)

Technology: Vacuum Tubes

The first generation computers used vacuum tubes for processing data and magnetic drums for memory. These computers were very large, consumed a lot of electricity, and produced excessive heat.

Characteristics

- Used vacuum tubes
- Very large in size
- High power consumption
- Produced much heat
- Slow compared to modern computers
- Machine language was used

Examples

- ENIAC
- EDVAC
- UNIVAC-I

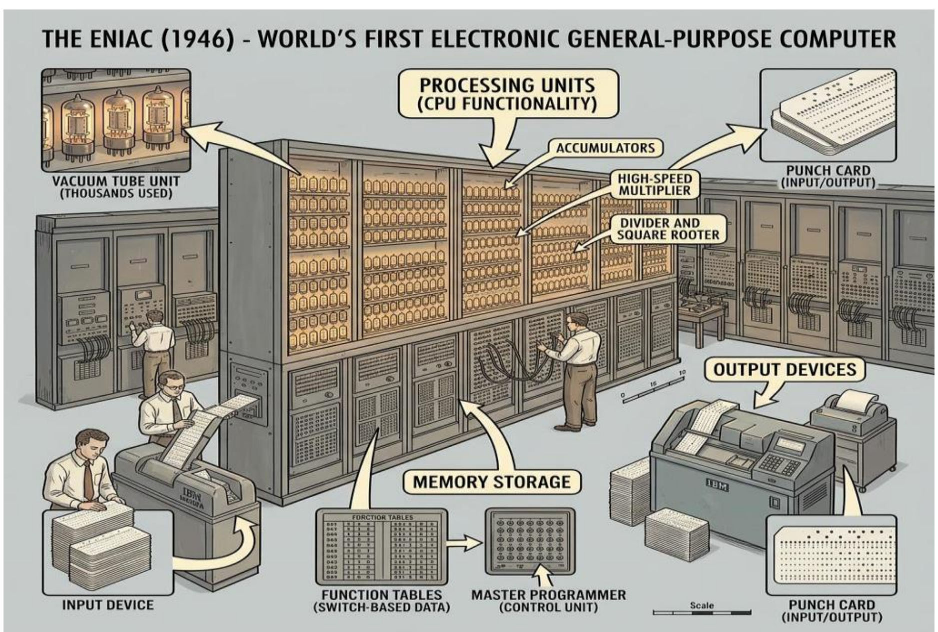

Fig 1.2.a) First Generation of Computers

2. Second Generation (1956–1963)

Technology: Transistors

The **second generation computers** replaced vacuum tubes with **transistors**, making computers smaller, faster, more reliable, and more energy-efficient.

Characteristics

- Used transistors
- Smaller than first generation
- Less heat generation
- Faster processing
- More reliable
- Assembly language used

Examples

- IBM 1401
- IBM 7094
- CDC 1604

Fig 1.2.b) Second Generation of Computers

3. Third Generation (1964–1971)

Technology: Integrated Circuits (ICs)

The **third generation computers** used **Integrated Circuits (ICs)** instead of transistors. This greatly increased speed and reduced the size and cost of computers.

Characteristics

- Used Integrated Circuits (ICs)
- Smaller and faster
- More reliable
- Lower power consumption
- Supported operating systems
- High-level programming languages

Examples

- IBM System/360
- PDP-8
- Honeywell-6000

Fig 1.2.c) Third Generation of Computers

4. Fourth Generation (1971–Present)

Technology: Microprocessors

The **fourth generation computers** use **microprocessors**, where the CPU is placed on a single chip. These computers are very fast, compact, affordable, and widely used today.

Characteristics

- Uses microprocessors
- Very high speed
- Small size
- Large memory
- Low cost
- User-friendly

Examples

- Desktop Computer
- Laptop
- Personal Computer (PC)

Fig 1.2.d) Fourth Generation of Computers

5. Fifth Generation (Present & Future)

Technology: Artificial Intelligence (AI)

The **fifth generation computers** are based on **Artificial Intelligence (AI)**. They are designed to learn, reason, understand natural language, and solve complex problems.

Characteristics

- Uses Artificial Intelligence (AI)
- Machine Learning
- Voice Recognition
- Robotics

Examples

- AI Assistants (ChatGPT, Siri, Google Assistant)
- Self-driving Cars
- Supercomputers

Fig 1.2.e) Fifth Generation of Computers

1.3 Features of a Computer System

- ✓ Accepts data through input devices.
- ✓ Processes data at high speed.
- ✓ Stores large amounts of information.
- ✓ Produces accurate results.
- ✓ Can work continuously without getting tired.

2. Characteristics of a Computer

Characteristics of a computer are the unique features that enable it to perform tasks efficiently, such as speed, accuracy, diligence, automation, versatility, reliability, multitasking, and large storage capacity.

Easy to Remember:

S A D S A V R M I

Speed → **A**ccuracy → **D**iligence → **S**torage → **A**utomation → **V**ersatility → **R**eliability → **M**ultitasking → **I**ntelligence (No IQ)

Fig 2.a) Characteristics of Computers

1. Speed

- A computer performs millions or billions of calculations in a fraction of a second.
- It completes tasks much faster than humans.

Example: Calculating the marks of 10,000 students in a few seconds.

2. Accuracy

- A computer produces highly accurate results if the input data and program are correct.
- Errors usually occur due to incorrect input or faulty instructions (Garbage In, Garbage Out - GIGO).

Example: Calculating bank interest without mistakes.

3. Diligence

- A computer does not get tired or bored.
- It can perform the same task repeatedly with the same efficiency.

Example: Printing thousands of bills continuously.

4. Storage Capacity

- A computer can store a large amount of data.
- Stored data can be retrieved whenever required.

Example: Storing thousands of photos, videos, and documents.

5. Automation

- Once a program is started, the computer performs all operations automatically until the task is completed.

Example: Running a backup every night automatically.

6. Versatility

- A computer can perform many different types of tasks.
- It can be used for education, gaming, programming, banking, designing, and research.

Example: A laptop can be used to attend online classes, create presentations, and watch videos.

7. Reliability

- A computer provides consistent and dependable results.
- It can work continuously for long periods without failure.

Example: ATM machines work 24×7 with reliable performance.

8. Multitasking

- A computer can perform multiple tasks at the same time.

Example: Listening to music while typing a document and downloading files.

9. Large Memory

- Computers can store vast amounts of information in RAM and secondary storage devices like SSDs and hard disks.

Example: Saving years of student records in a database.

10. No Intelligence (No IQ)

- A computer cannot think or make decisions on its own.
- It only follows the instructions given by the user.

Example: It cannot solve a problem unless programmed to do so.

3) Applications of Computer Systems in Daily Life

a) Education

b) Banking

c) Health care

d) Business

e) Scintific Research

f) Entertainment

g) Communication

h) Government Services

a) **Education:** Computers support online learning, digital classrooms, exams, and educational research.

Fig 3.a) Computers in the educational sector

b) **Banking:** Computers manage customer accounts, online transactions, ATMs, and financial records securely.

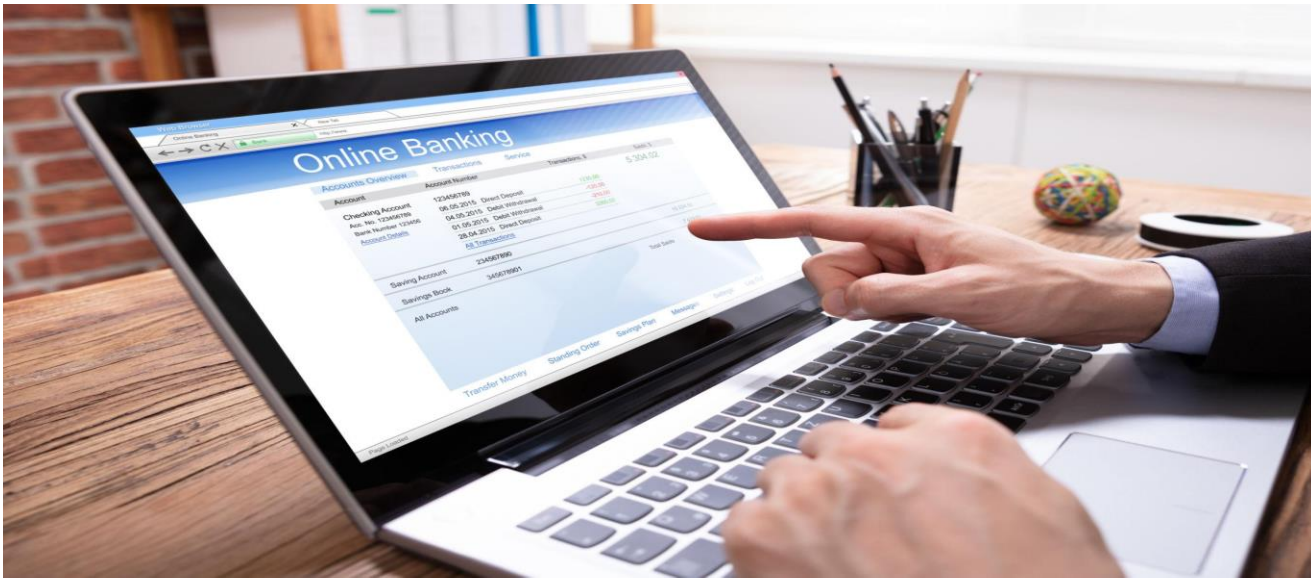

Fig 3.b) Computers in Banking sector

c) Healthcare: Computers help maintain patient records, perform medical diagnosis, and support hospital management.

Fig 3.c) Computers in Health care sector

d) **Business:** Computers are used for accounting, inventory management, communication, and decision-making.

Fig 3.d) Computers in Business sector

e) **Scientific Research:** Computers analyze large datasets, perform simulations, and assist in scientific discoveries.

Fig 3.e) Computers in Scientific Research sector

f) **Entertainment:** Computers provide games, movies, music, video streaming, and digital content creation.

Fig 3.f) Computers in the Entertainment sector

g) **Communication:** Computers enable email, video conferencing, social media, and instant messaging worldwide.

Fig 3.g) Computers in the Communication sector

h) **Government Services:** Computers help deliver online public services, maintain citizen records, and improve administrative efficiency.

Fig 3.h) Computers in Government sector

4. Basic Components of a Computer System

A computer system consists of several interconnected components that work together.

Block Diagram

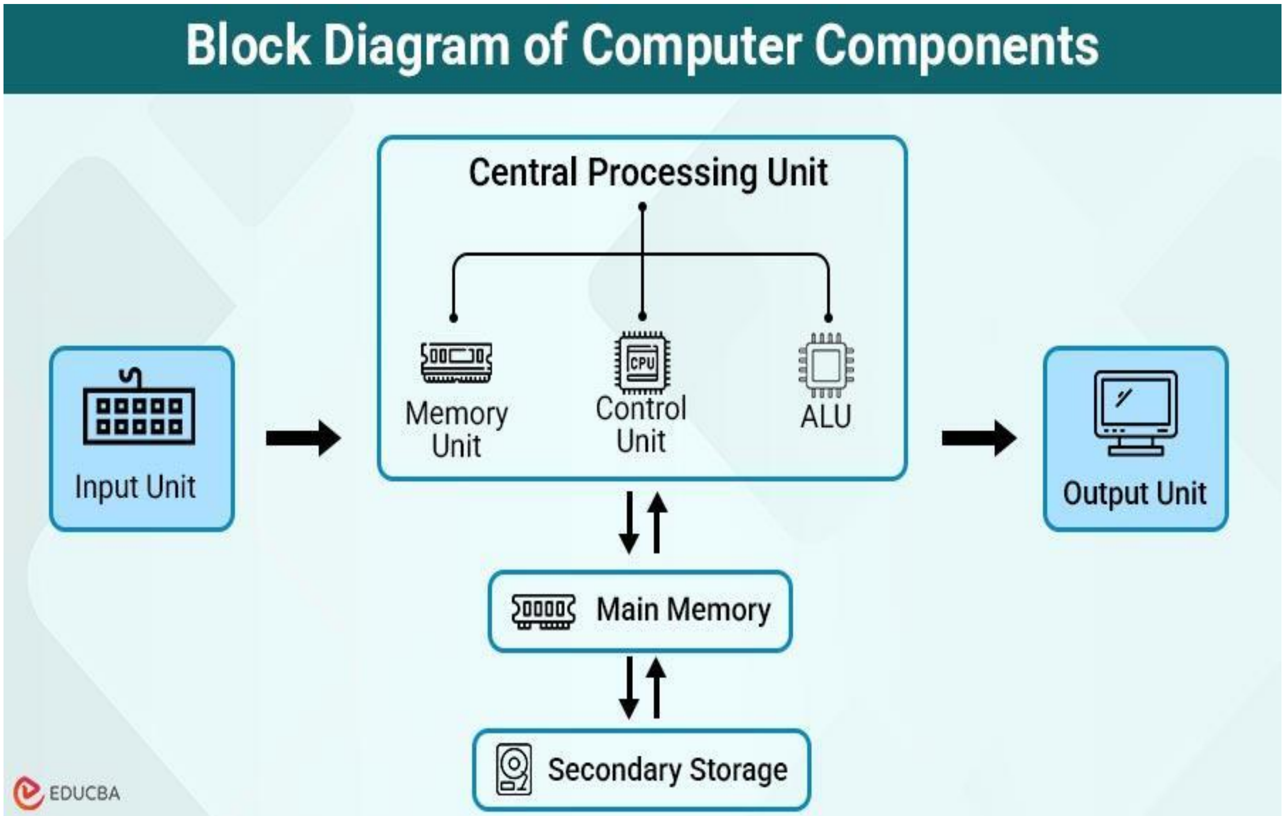

Fig 4.a) Block Diagram of a Computer

4.1. Input Unit

Input devices are hardware components used to enter data and instructions into a computer. They help users communicate with the computer.

Common Input Devices

- Keyboard
- Mouse
- Scanner
- Microphone

- Webcam
- Joystick
- Barcode Reader

Functions

- Accepts data.
- Converts input into machine-readable form.
- Sends data to the CPU.

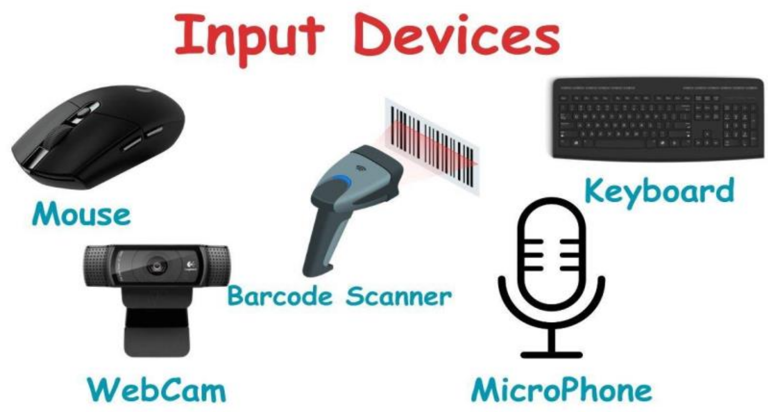

Fig 4.b) Input Devices

a)Keyboard

The keyboard is the most commonly used input device. It consists of alphabet keys, number keys, function keys, and special keys. It is mainly used for typing documents, entering data, and giving commands to the computer.

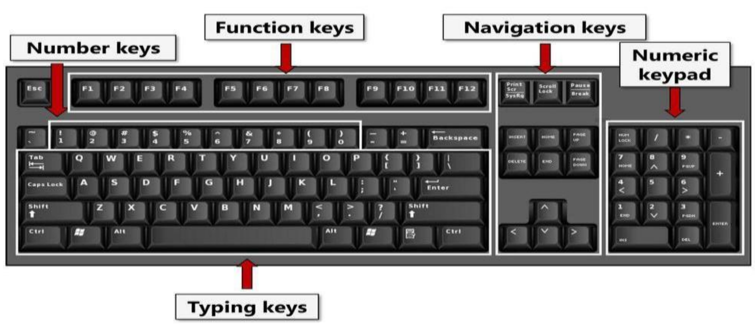

Fig 4.c) Keyboard

b)Mouse

A mouse is a pointing device used to move the cursor on the screen. It helps users click, drag, scroll, and select files, folders, and other objects quickly.

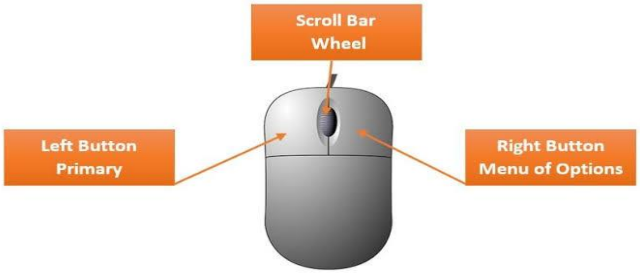

Fig 4.d) Mouse

c)Scanner

A scanner converts printed documents, photographs, and images into digital format. It is commonly used in offices, schools, and hospitals for storing documents electronically.

Fig 4.e) Scanner

d)Microphone

A microphone captures voice and sound and sends them to the computer. It is used for voice recording, online meetings, video conferencing, and speech recognition.

Fig 4.f) Microphone

4.2. Central Processing Unit (CPU)

The CPU is known as the **Brain of the Computer**.

It performs all calculations and controls the operation of the computer.

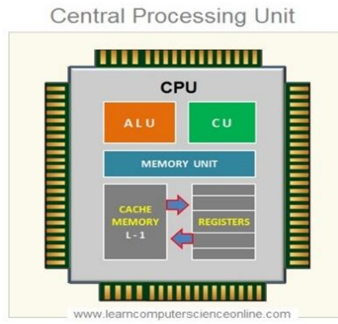

Fig 4.g) CPU

Components of CPU

a) Arithmetic Logic Unit (ALU)

The Arithmetic Logic Unit (ALU) is a part of the CPU that performs all arithmetic and logical operations. It is called the calculator of the computer because it carries out mathematical calculations and comparisons.

ALU Does

- ➕ Addition
- ➖ Subtraction
- ✖ Multiplication
- ➗ Division
- 🔍 Compares values (>, <, =)
- ✔ Makes logical decisions (AND, OR, NOT)

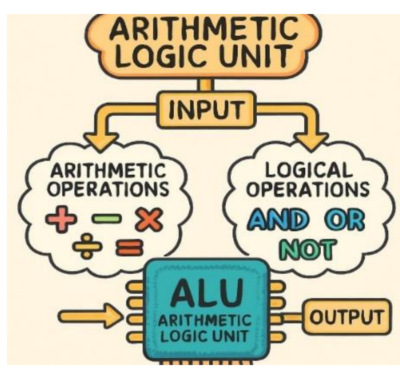

Fig 4.h) ALU

b) Control Unit (CU)

The **Control Unit (CU)** is a component of the **Central Processing Unit (CPU)** that **controls and coordinates all operations of the computer**. It manages the flow of data and instructions between the CPU, memory, and input/output devices.

Functions

- Fetches instructions from memory.
- Decodes the instructions.
- Controls the execution of instructions.
- Coordinates communication between CPU, memory, and I/O devices.
- Manages the sequence of all computer operations.

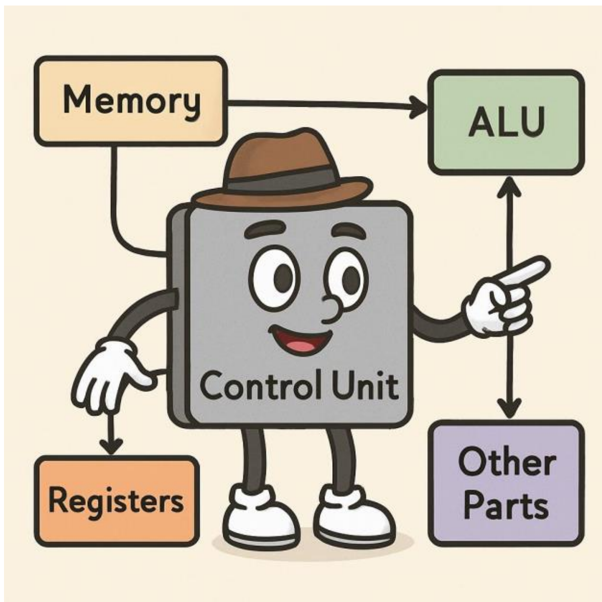

Fig 4.i) Control Unit

4.3) Memory Unit (MU)

The **Memory Unit** is a part of the computer that **stores data, instructions, and processing results**. It holds information temporarily or permanently so that the CPU can access it whenever needed.

Functions

- Stores input data and program instructions.
- Stores intermediate processing results.
- Stores the final output before it is displayed.
- Provides data to the CPU whenever required.

Types of Memory

1. Primary Memory (Main Memory)

- It is directly accessed by the CPU.
- It is fast but has limited storage capacity.
- It is used to store data and programs currently in use.

Examples:

- **RAM (Random Access Memory):** Temporary memory. Data is lost when the computer is turned off.
- **ROM (Read Only Memory):** Permanent memory. Stores important startup instructions.

2. Secondary Memory

- It stores data permanently.
- It has a larger storage capacity than primary memory.
- It is slower than primary memory.

Examples:

- Hard Disk
- SSD (Solid State Drive)
- Pen Drive (USB Flash Drive)
- CD/DVD
- Memory Card

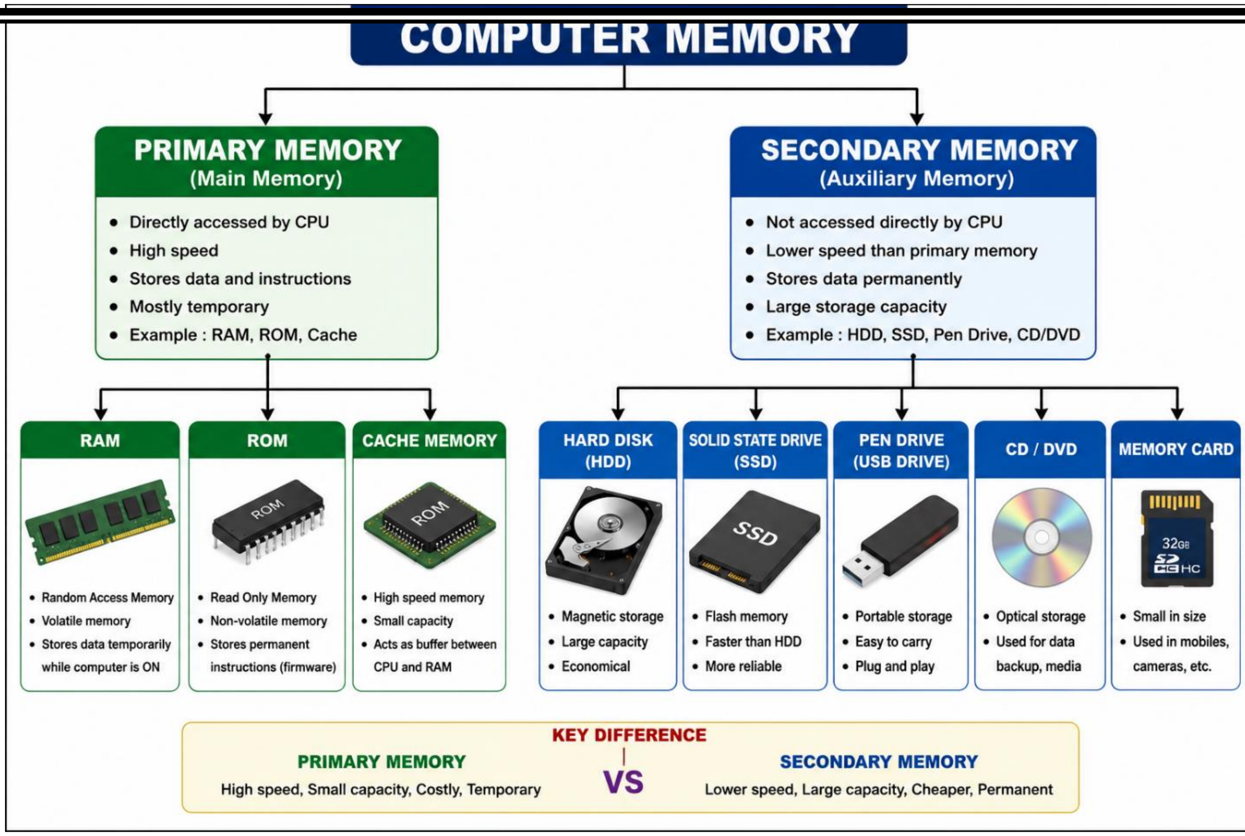

Fig 4.j) Types of Memory

4.4. Output Unit

Output devices are hardware components that display or produce the results of the data processed by the computer. They convert digital information into a form that users can see, hear, or use.

Output Devices

- Monitor
- Printer
- Speaker
- Projector
- Plotter

Functions:

- Receives processed data.
- Converts binary information into human-readable form.
- Presents results to the user.

a)Monitor

A monitor is the most common output device. It displays text, images, graphics, videos, and application results on the screen. It is also called the Visual Display Unit (VDU).

Fig 4.k) Scanner

b)Printer

A printer produces a hard copy of documents, reports, and images on paper. Printers are commonly used in schools, offices, and businesses.

Fig 4.l) Printer

c)Speakers

Speakers convert digital audio signals into sound. They are used to play music, videos, voice recordings, alarms, and system notifications.

Fig 4.m) Scanner

d)Projector

A projector displays the computer's screen on a large surface, making it suitable for classrooms, seminars, business meetings, and presentations.

Fig 4.n) Scanner

Other Devices

Computer Cabinet

A computer cabinet is an enclosure with fitted, fixed or removable side panels and doors. The cabinet contains a computer rack for mounting computers or other electronic equipment. Cabinets come in a variety of sizes, colors and styles.

Fig 4.o) Scanner

CPU Heat Sink & Fan

Computer cooling is required to remove the waste heat produced by computer components. heatsink with fan clipped onto a microprocessor. heatsinks cooled by airflow with Computer fans reduces the temperature by actively exhausting hot air.

Fig 4.p) Scanner

Microprocessors

A Microprocessor is a programmable electronic integrated circuit (IC) that acts as the brain of a computer. It processes data, executes instructions, performs arithmetic and logical operations, and controls all the functions of the computer.

Features of a Microprocessor

- It is a single integrated circuit (IC).
- It performs millions or billions of operations per second.
- It executes program instructions stored in memory.
- It controls communication between hardware components.
- It is small, fast, and energy-efficient.

Types of Microprocessors

Type	Description	Examples
4-bit Microprocessor	Processes 4 bits of data at a time. Used in simple calculators and basic embedded systems.	Intel 4004
8-bit Microprocessor	Processes 8 bits of data at a time. Used in early computers and microcontroller-based devices.	Intel 8085, Zilog Z80
16-bit Microprocessor	Processes 16 bits of data at a time. Faster than 8-bit processors and used in personal computers.	Intel 8086, Intel 8088
32-bit Microprocessor	Processes 32 bits of data at a time. Supports multitasking and modern operating systems.	Intel Pentium, ARM Cortex-A
64-bit Microprocessor	Processes 64 bits of data at a time. Provides high speed and large memory support. Used in today's computers, laptops, and smartphones.	Intel Core i5/i7/i9, AMD Ryzen

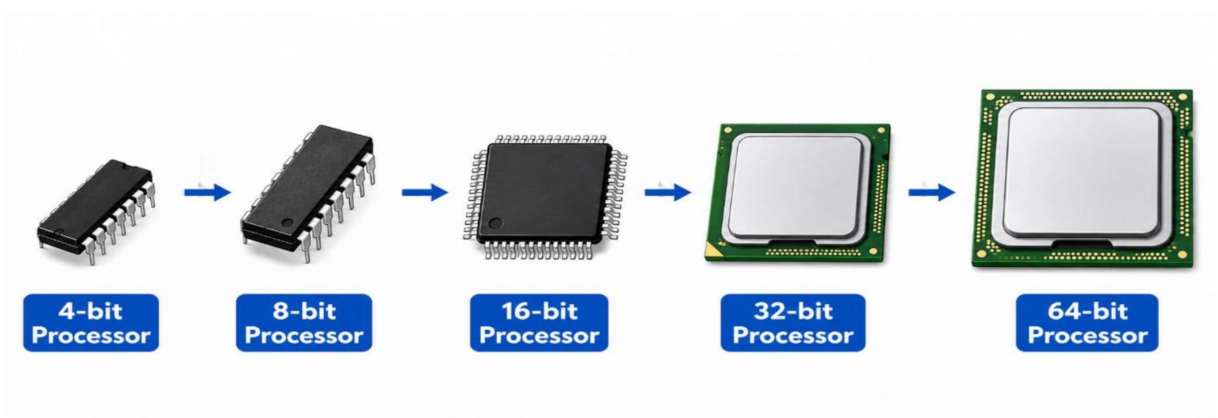

Fig 4. q) Types of Micro Processors

5)Advantages and Limitations of Computers

- High speed
- High accuracy
- Large storage capacity
- Reliability
- Automation
- Multitasking
- Easy communication
- Efficient data processing

Limitations of Computers

- Cannot think independently.
- No emotions or intelligence.
- Dependent on electricity.
- Vulnerable to viruses and cyberattacks.
- Requires proper instructions to function.

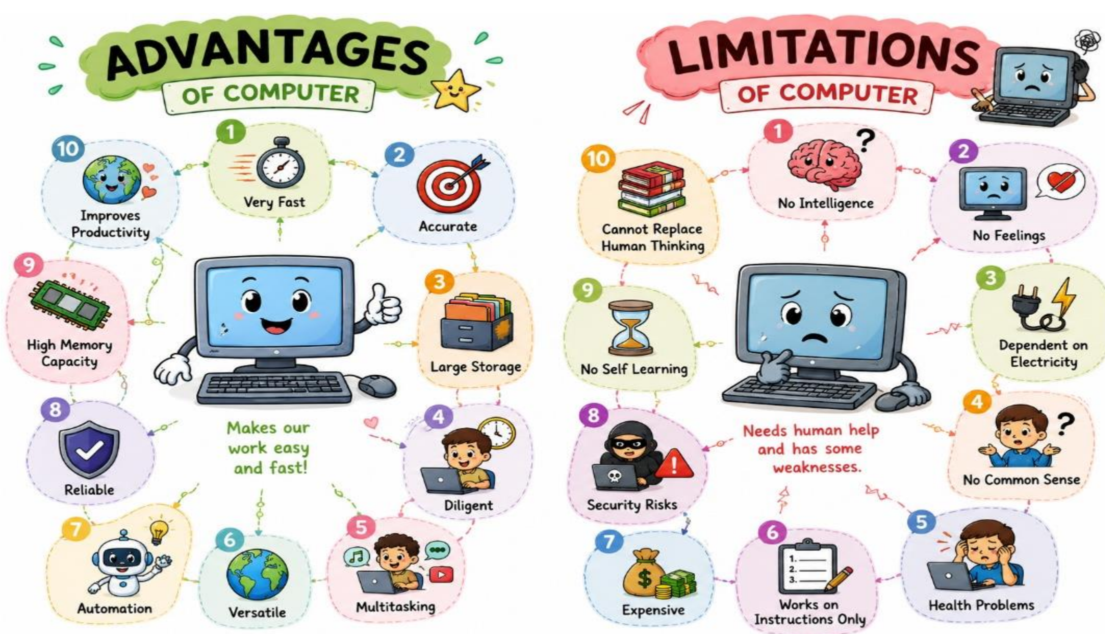

Fig 5.a) Advantages and Limitations of Computer

MCQs

1. A computer is an electronic device that:

A) Only stores data
B) Only prints documents
C) Accepts data, processes it, and produces output
D) Only plays games

2. Which cycle represents the basic working of a computer?

A) Input → Storage → Output
B) Process → Input → Output
C) Input → Process → Output
D) Output → Input → Process

3. Which of the following is an input device?

A) Printer
B) Monitor
C) Keyboard
D) Speaker

4. Which device is mainly used to point and click on the screen?

A) Scanner
B) Mouse
C) Keyboard
D) Printer

5. A scanner is used to:

A) Print documents
B) Record sound
C) Convert paper documents into digital form
D) Display videos

6. Which input device captures voice?

A) Monitor
B) Printer
C) Microphone
D) Projector

7. Which of the following is an output device?

A) Mouse
B) Keyboard
C) Monitor
D) Scanner

8. Which device produces a hard copy?

A) Monitor
B) Speaker
C) Printer
D) Scanner

9. Which output device converts digital signals into sound?

A) Monitor
B) Speaker
C) Projector
D) Mouse

10. The "brain of the computer" is:

A) RAM
B) CPU
C) Hard Disk
D) Keyboard

11. Which characteristic means a computer works very fast?

A) Storage
B) Reliability
C) Speed
D) Diligence

12. GIGO stands for:

A) Good Input Good Output
B) Garbage In Garbage Out
C) Great Input Great Output
D) General Input General Output

13. A computer can work continuously without getting tired. This characteristic is called:

A) Accuracy
B) Speed
C) Diligence
D) Storage

14. Which characteristic allows a computer to perform many different tasks?

A) Reliability
B) Versatility
C) Automation
D) Memory

15. Which characteristic means the computer performs tasks automatically after starting a program?

A) Accuracy
B) Automation
C) Diligence
D) Reliability

16. A computer cannot think on its own because it has:

A) High speed
B) No Intelligence
C) Reliability
D) Automation

17. Which sector uses computers for online learning?

A) Banking
B) Education
C) Entertainment
D) Business

18. ATMs mainly belong to which application area?

A) Education
B) Banking
C) Healthcare
D) Research

19. Hospitals use computers mainly for:

A) Playing games
B) Medical diagnosis and patient records
C) Watching movies
D) Music

20. Which unit accepts data from the user?

A) Output Unit
B) CPU
C) Input Unit
D) Memory Unit

21. ALU stands for:

A) Arithmetic Logic Unit
B) Automatic Logic Unit
C) Arithmetic Language Unit
D) Automatic Learning Unit

22. Which component performs arithmetic calculations?

A) CU
B) RAM
C) ALU
D) ROM

23. Which CPU component controls all computer operations?

A) ALU
B) ROM
C) Control Unit
D) Hard Disk

24. Which memory loses data when power is switched off?

A) ROM
B) RAM
C) Hard Disk
D) DVD

25. Which memory stores startup instructions permanently?

A) RAM
B) SSD
C) ROM
D) Pen Drive

26. Which is an example of secondary memory?

A) RAM
B) ROM
C) SSD
D) Cache

27. Which memory has larger storage capacity?

A) Primary Memory
B) Secondary Memory
C) Cache Memory
D) Registers

28. Which unit displays processed information?

A) Input Unit
B) Memory Unit
C) Output Unit
D) CPU

29. Which of the following is an advantage of computers?

A) They have emotions.
B) They think independently.
C) They perform tasks with high speed and accuracy.
D) They never require electricity.

30. Which of the following is a limitation of computers?

A) High speed
B) Large storage
C) Reliability
D) Cannot think independently

Review Questions

1. Define a computer system and list its basic components.
2. Explain characteristics of a computer.
3. Write short notes on the applications of computers in daily life.
4. Explain the functions of the CPU, ALU, and Control Unit.
5. Differentiate between Primary Memory and Secondary Memory with examples.